

LAB 1 – USE-CASE FLOW SPECIFICATION

UC-02: Grant Time-Bounded Consent

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Use Case

UC-02 – Grant Consent

Primary Actor: Patient

Supporting Actor: Verified Clinic Doctor

Preconditions

1. The patient is authenticated.
2. The requested doctor has been verified by a Clinic Administrator.
3. The requested diagnostic records exist in the patient's health record.

Postconditions

1. A consent permission is stored with the selected doctor, selected records, grant time, and expiry time.
2. The doctor can access only the selected records until the expiry time.
3. The consent-grant event is written to the append-only audit trail.

Main Success Scenario

1. Patient opens Consent Management.
2. System displays available diagnostic records and eligible verified clinic doctors.
3. Patient selects one or more diagnostic records.
4. Patient selects a verified clinic doctor.
5. Patient specifies the consent duration/expiry time.
6. System validates the doctor, record scope, and expiry details.
7. System creates the consent permission and marks it Active.
8. System records the grant event in the append-only audit trail.
9. System displays a confirmation containing the recipient, records, and expiry time.
10. Use case ends successfully.

Alternate Flow – Invalid/Expired Consent Details

- 6a. System detects that the selected doctor is no longer verified or the expiry time is invalid.
 - 6a1. System rejects the consent request.
 - 6a2. System displays the reason for rejection and does not create an active consent.
 - 6a3. Patient may correct the details and resubmit.
 - 6a4. If the patient cancels, the use case ends without creating consent.

Traceability: This flow operationalizes FR-001 and the scenario's time-bounded, patient-controlled consent model.